

3D-PRINTED GEOMETRY-TUNABLE MICROFLUIDIC ELECTROPORATION FOR HIGH-THROUGHPUT BACTERIAL TRANSFORMATION

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Motivation

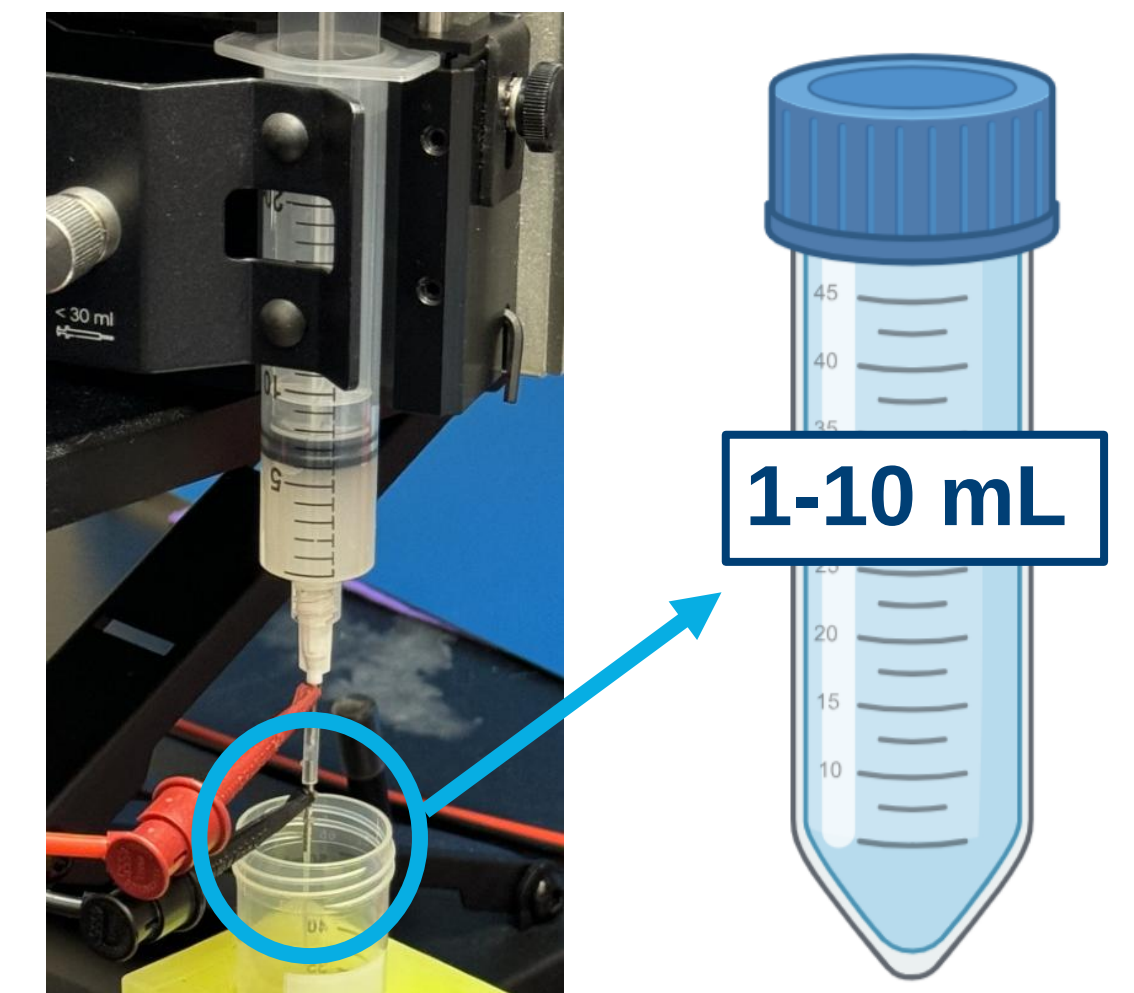
Bacterial electroporation is essential for genome engineering, strain development, and synthetic biology, but most transformations are performed at microliter-scale volumes, limiting throughput and reproducibility.

- **Cuvette-based electroporation:** A manual, operator-dependent process with limited-volume (<100 μL) $\rightarrow$ **time-consuming, operator-dependent outcome and limited applications.**
- **Flow-through electroporation [1]:** A process address volume limitations but suffers from variation in manual device assembly and limited field strength $\rightarrow$ **operator-dependent outcome and limited applications.**

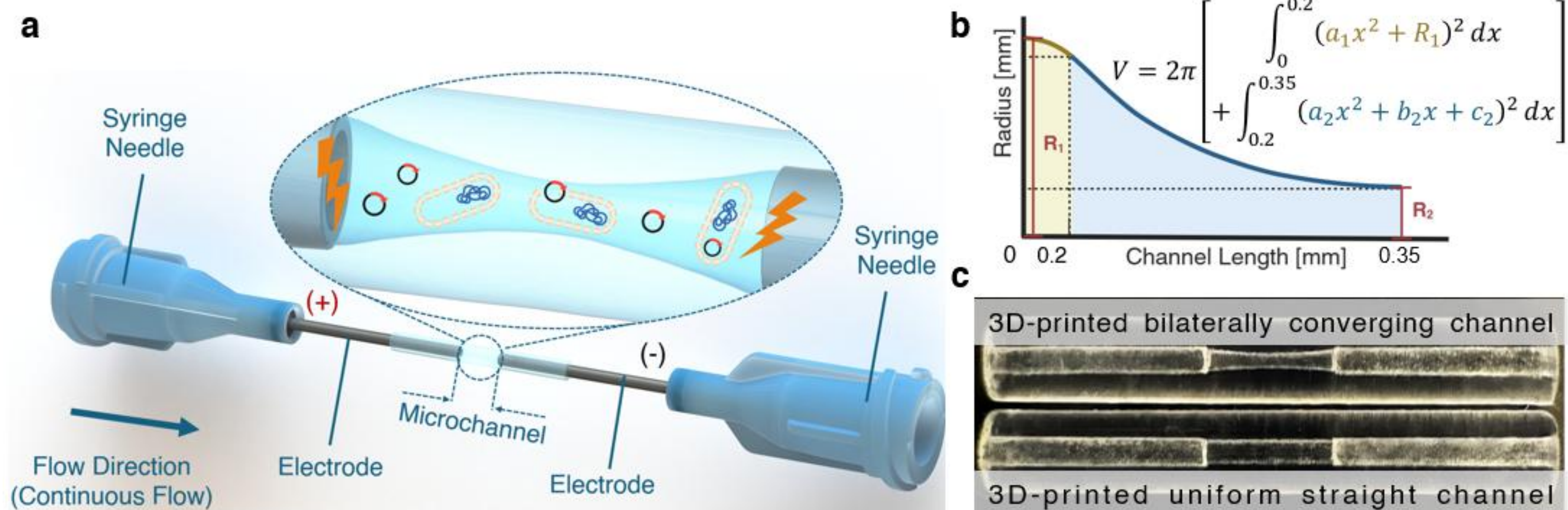
Cuvette-based Electroporation



Microfluidic Electroporation

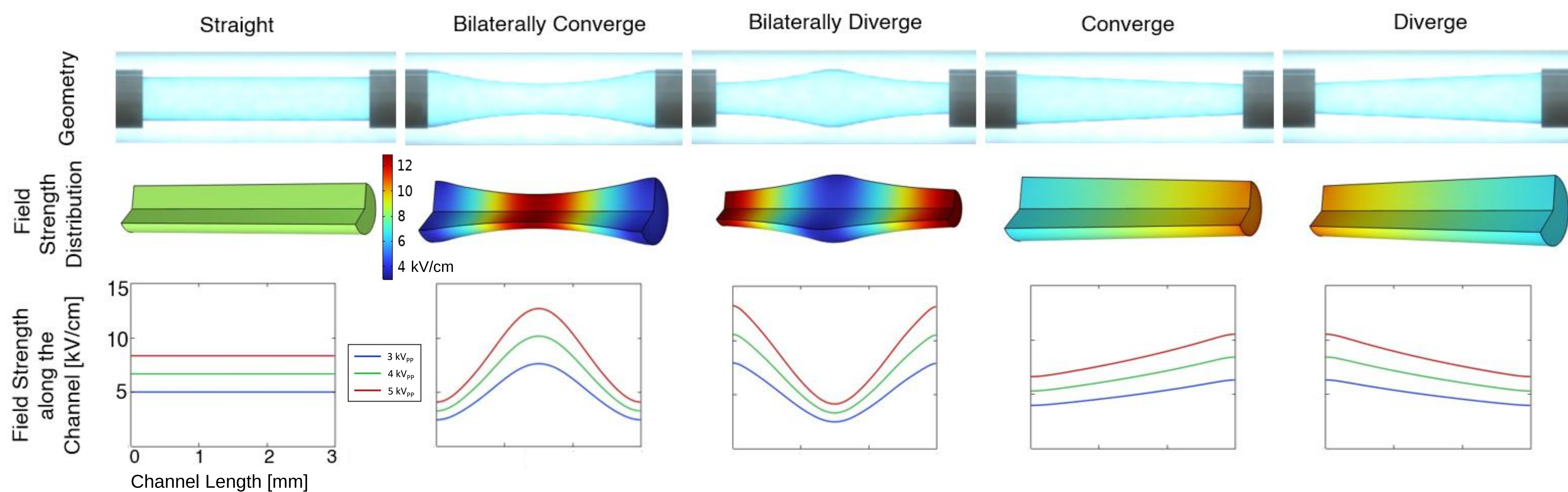


3D-Printed Flow-Through Electroporation Device with Tunable Geometry



- **Two syringe-needle electrodes** connected by a **3D-printed microchannel** as the electroporation chamber
- **Constant volume and length** maintained across designs
- 3D printing enables rapid iteration and **consistent device fabrication**

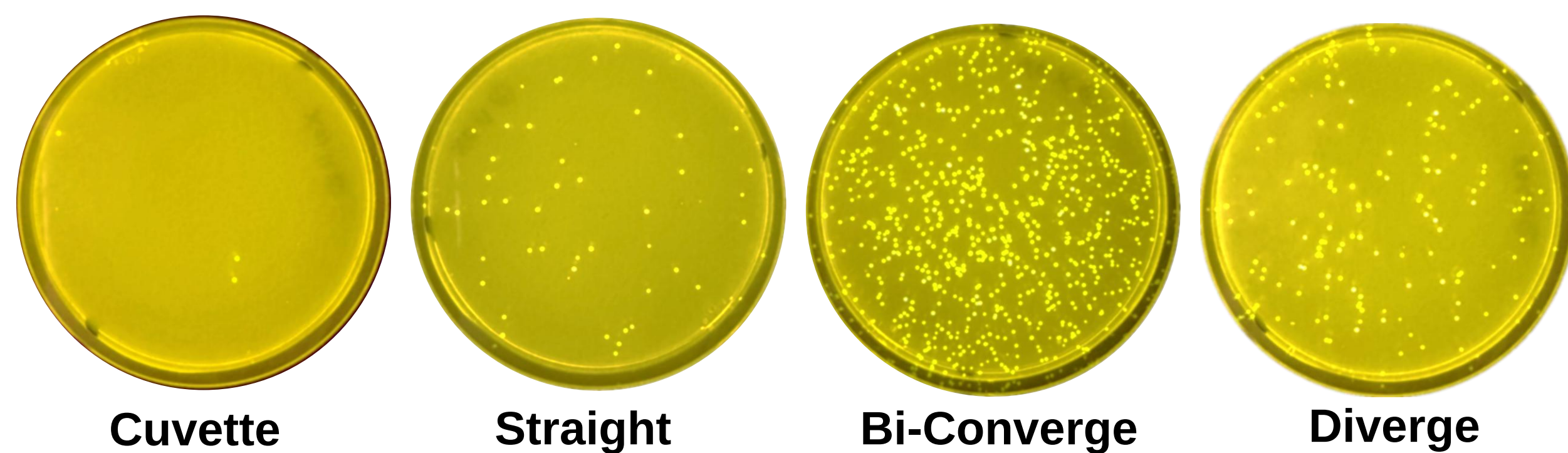
Simulated Field Distribution for Different Geometries



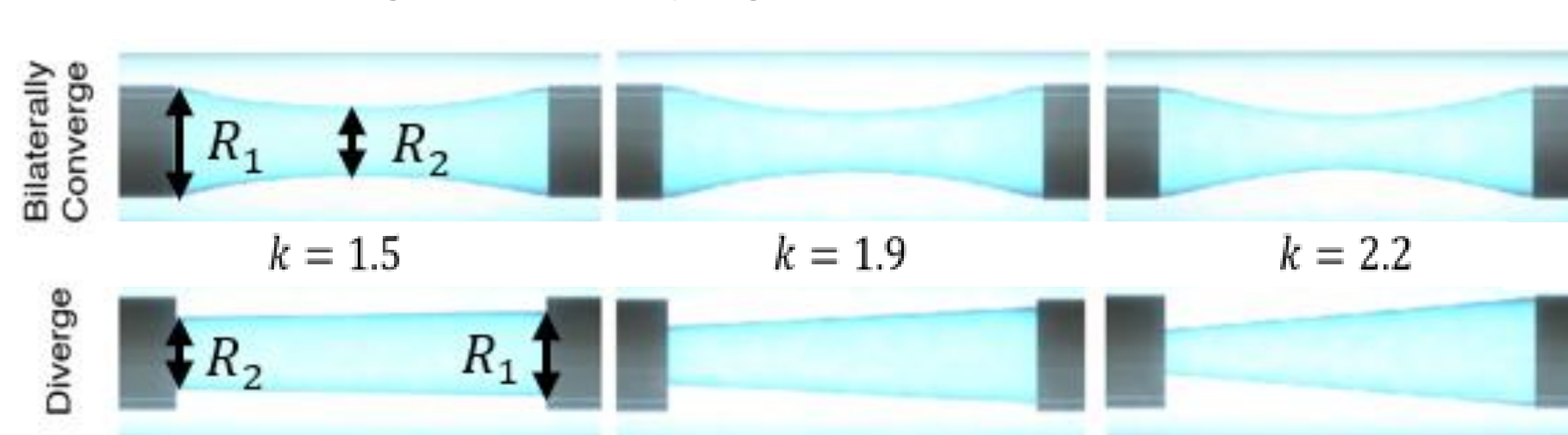
- The design ensures **smooth transition** across regions of varying electric field strength.
- **Non-uniform geometries locally amplify electric field intensity.**
- Bilaterally converging and diverging designs exhibit up to **55% higher field strength** than the straight channel at the **same applied voltage.**

Optimization of Constriction Ratio for Maximum Yield

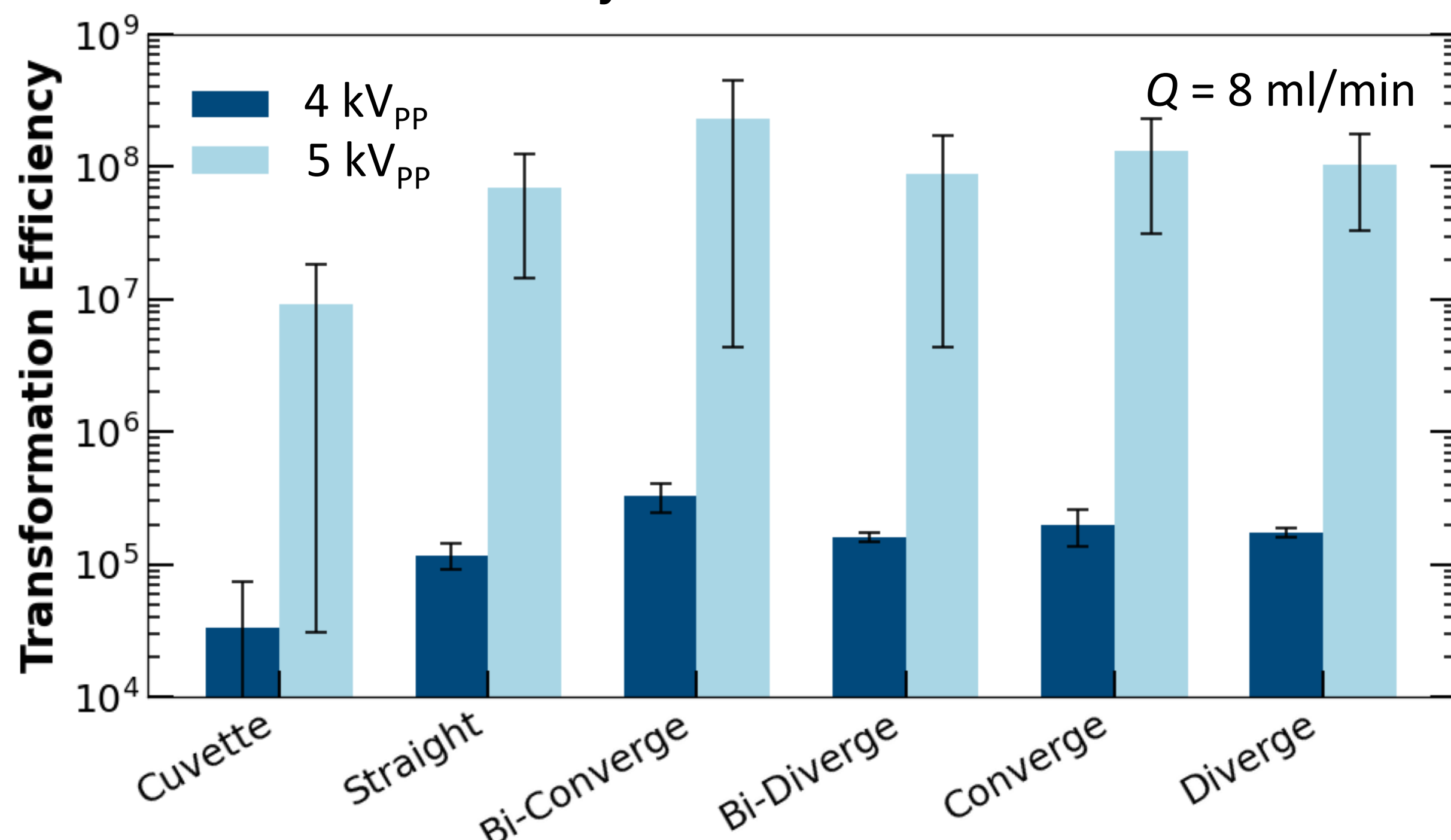
Representative Fluorescent Plates for Different Geometries



Channel Designs with Varying Constriction Ratios $k = R_1 / R_2$

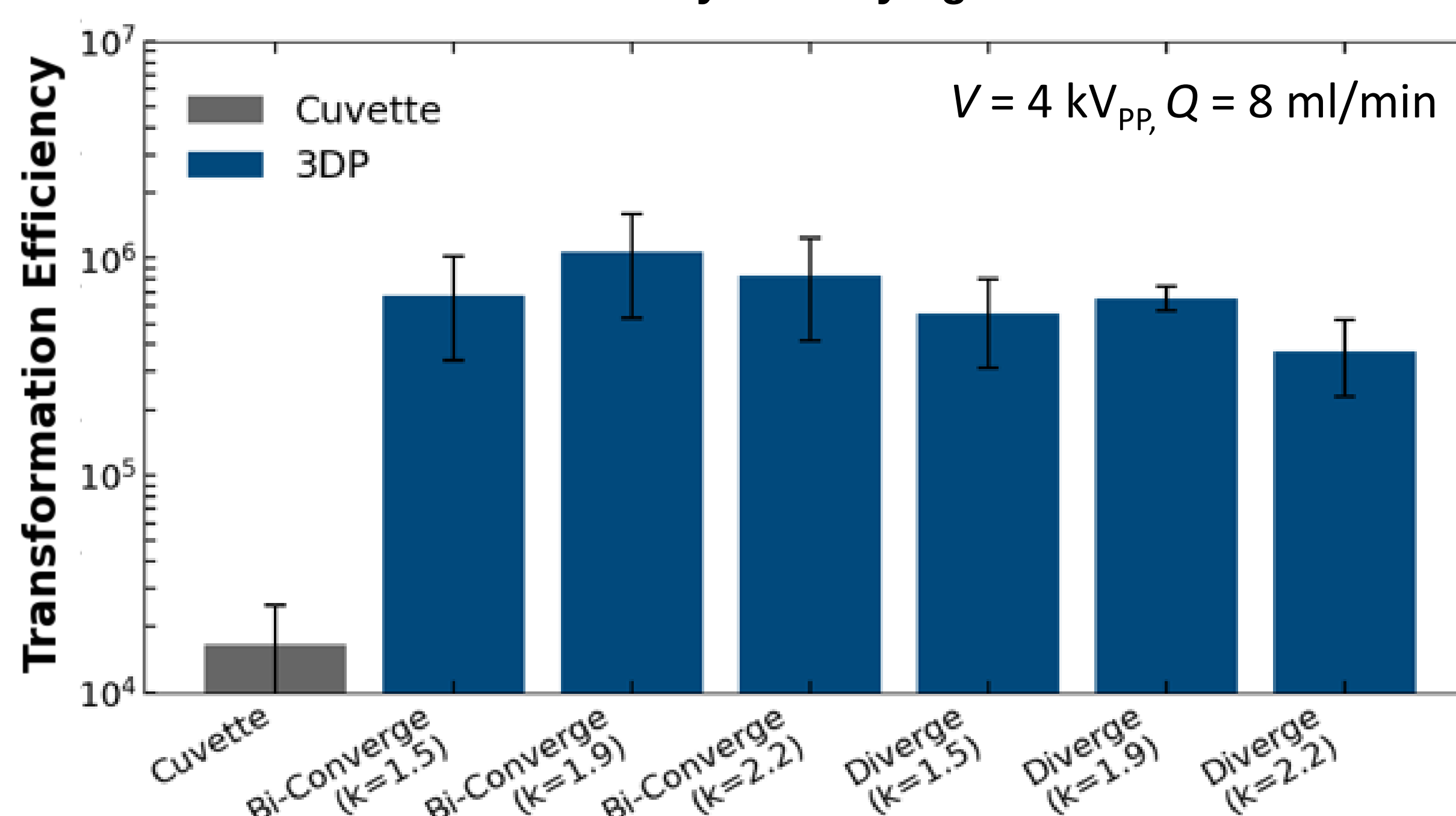


Transformation Efficiency Across Different Channel Geometries



- **Bilateral convergence performance surpassed both cuvette and other 3D-printed designs**

Transformation Efficiency for Varying Constriction Ratios



- **$k = 1.9$ achieved the highest transformation efficiency** in both geometries

Conclusions

- The **3D-printed, geometry-tunable microfluidic device** enables **reproducible and scalable electroporation.**
- **Localized electric-field enhancement** contributes to improved transformation efficiency.
- Identifies geometries that **maximize field strength and transformation efficiency.**
- This method establishes a **flexible and high-volume platform** for efficient bacterial electroporation.

References

[1] P.-H. Huang et al., PLOS Biology, vol. 20, no. 9, Sep. 2022.

Acknowledgement

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